

Yi Zheng TEOH (Ethan)

Software Engineer with the passion for synergising technology with users.

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EDUCATION

University College London

Candidate for Masters of Science in Systems Engineering Management

Concentration: Systems Engineering and Project Management

Coursework: Systems Thinking Engineering Management, Business Environment, Project Management

London, United Kingdom

Graduation: December 2025

Cumulative Score: 65.2%

National University of Singapore

Bachelor of Engineering in Computer Engineering with 2nd Major in Design and Innovation

Concentration: Firmware and Software Development

Coursework: Power Electronics, IoT, UI/UX, VR/AR development, Human-Robot Interface modeling

Research: Racecar Operating System, Driver controls, Power Distribution(FSAE), Optimised Mass EV Charging Control

Singapore

Graduation: July 2023

Cumulative Score: 70.4%

Ngee Ann Polytechnic

Diploma in Engineering Science

Concentration: Automation and Mechatronic systems

Research: Autonomous Wheelchair, Stereo 2D LiDAR for 3D mapping.

Singapore

Graduation: May 2018

Cumulative Score: 93.3%

WORK EXPERIENCE

SNAPPIT (BIGTAM)

Software Lead (Part-time)

London, United Kingdom

February 2025 – Present

- Designed architecture and developed the platform from the ground up including workflow..
- Development full software stack: react native mobile app, android SDK (Kotlin), Nextjs web app, Supabase, GCP.
- Setup/Developed custom Auth service, DB schema/RPC(PostgreSQL), REST services with nodejs, Deno.
- Implemented CiCD pipeline for application using Github Actions and Fastlane, backend with docker and Github actions, managing env secrets, encoding with base64, encryption protocols using (Blowfish, MD5, and SHA1).

ION MOBILITY

Systems Test Engineer/Technical Project Manager, Systems (Embedded, Electronics, Performance)

Singapore/Indonesia

January 2022 – September 2024

- Continuous Testing: I built CI/CD pipelines (Terraform, GitHub Actions, SSH) and automated hardware-in-the-loop test setup in Golang to improve the reliability of real-time control and IoT systems (ESP, Quectel LTE/GPS). This required me to write the test-cases to run on the HiL testbench.
- Diagnostics: I initiated and developed an integrated diagnostics tool in Rust to help engineers and technicians identify issues across OS, networking, and timing-critical logic. This evolved into an automated service tool (Rust for interface control, Python for running fault tree-analysis analysis/traversing logic) used in post-deployment servicing.
- Production Architecture: I designed the architecture to manage key and firmware data flow within the factory, leading the development of the individual tools and coordinating with production to test feasibility and speed of operation. (This included writing the proposal with user and technical requirements/breakdown, design, and integration testing.)
- Reporting: I created a log-based reporting framework that ensured traceability and clear routing of issues to relevant stakeholders, integrating seamlessly with our diagnostics stack.
- Workflows: I brought my teams onto Jira, adopting an Agile framework with tailored Scrum cycle to include integration, verification, and validation stages, strengthening delivery while accelerating bug resolution.
- Developed company website with backend integration (NextJS, NodeJS, PostgreSQL, OAUTH 2.0)
- Developed BMS firmware (S32K144, Simulink), VCU firmware (S32K118 RTOS, ESP32, Nordic 52834).
- Led team of 8 engineers planned timeline, assigned tasks, and implemented task tracking and adopted agile lifecycle to enhance software features delivery rate by more than 300%, completing production ready firmware/software suite 8 months for an exhibition, doubling motorcycle pre-orders, and an investor showcase (Showcase led to buyout)
- Designed systems related factory production processes based on APQP adopting PFMEA to achieve 1.5min takt time
- Technical representative at exhibitions and core sales events, conducted training for sales representatives regarding vehicle features and operation.

MINISTRY OF DEFENCE SINGAPORE*Platoon Commander, Artillery HIMARS***Singapore***August 2018 – August 2020*

- I created the SOP for adopting a new communication system improving speed, security, and reliability
- Developed an application for safety reporting that reduced reporting time by 60%
- Led a platoon to achieve cat A (Highest performance result awarded) for the team's first operational exercise

SHIRO-WA*Project Engineer***Singapore***February 2018 – June 2018*

- I developed a mobile app, experimented using Java initially, however quickly changed to kotlin for runtime robustness
- Implemented a basic pipeline for deployment
- Designed SOP for a new business venture and designed the packages too using Adobe Photoshop and Illustrator

MORGANFIELDS*Waiter/ Bartender (Part-time)***Singapore***June 2016 – August 2018*

- Conducted sales service and preparation of food and beverages
- Actively contributed to achieve daily sales target of SGD10K

XMATEC*Sales Assistant***Singapore***January 2015 – February 2015*

- Carried out cold selling of Minolta copiers. Prior to starting my diploma

NATIONAL HEALTHCARE GROUP*IT Assistant***Singapore***October 2014 – January 2015*

- Documented all drugs and supported the build of a database of drugs carried by the organisation for a new automated dispensing system

LEADERSHIP

NATIONAL UNIVERSITY OF SINGAPORE, SCHOOL OF COMPUTING CLUB*Project Director, Academic Department***Singapore***September 2020 – September 2021*

- Led a team to organise orientations for more than 500 freshmen, including mock test experiences and advice

NGEE ANN POLYTECHNIC, NAVIGATORS CLUB*President***Singapore***January 2017 – January 2018*

- Led a group to plan and run activities throughout the year

SKILLS, INTERESTS & AWARDS

Languages: Native English and Chinese speaker**Software Expertise:** C++, C, Golang, Java, Python, CSS (TailwindCSS), TypeScript(React Native, NextJs, Deno), JavaScript (NodeJs, Express), PostgreSQL(Schema, query, RPC), HTML**Software Experience:** Assembly, basic, Rust, Verilog, C#, Python (NumPy, Matplotlib, Panda, SciPy, TensorFlow, PyTorch), Dart, Kotlin, Kafka, Swift, NoSQL, Redis**Testing:** Hardware-In-Loop, Software-In-Loop, Jest, Selenium, Cucumber.**Frameworks:** Next.js, React Native, Flutter, Express.js, Spring Boot**Tools/Platforms:** Cloudflare CircleCI, AWS, GCP, Github (Actions, Pages, Secrets), Mender, Docker.**Development Methodology:** Agile, SCRUM, TDD, BDD, Waterfall, V-model.**Others:** Model-based Design (Matlab, Simulink, test harness), CAD (Solidworks, Inventor, AutoCAD), Electrical (Altium, KiCAD), Robot Operating System, Real-Time Operating System, Networking (OSI), Windows Powershell, Linux Bash.**Interests:** UI/UX design, finance, marketing strategy, fintech, photo editing, video editing, animation, battery technology**Awards:** Venture Initiation Program Award (2022 Winner), James Dyson Award (2021 National Runner Up)